

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: the website crashed is due to overwhelming traffic to the web page

The logs show that: 47 messages were sent and received by the web browser in 3.1 seconds. It shows that within this time frame multiple attempts are made by employee computers.

This event could be: IP spoofing

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. When a request is made by the user to visit the web page, a SYN packet is requested
2. The website responds with SYN, ACK packet, agreeing to the connection.
3. The final step is the ACK packet being acknowledged and being granted permission to connect. Then a successful TCP connection is established

Explain what happens when a malicious actor sends a large number of SYN packets all at once: It overwhelms the web page causing it to crash, therefore the usual TCP connections can not be carried out.

Explain what the logs indicate and how that affects the server: The logs indicate an overwhelming SYN packet request that overloads the system, customers attempting to connect will not be able to connect as the attack has caused an outage.